

Artificial Intelligence (AI)-Driven Design of Microbial Communities for Sustainable Agriculture and Climate Resilience

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1 Introduction

Nitrogen fertilizers are responsible for over 50% of anthropogenic nitrous oxide (N_2O) emissions [1]. N_2O is a potent and long-lived greenhouse gas, with a global warming potential nearly 300 times greater than carbon dioxide (CO_2), and it is the most significant ozone-depleting substance in the stratosphere [2]. Microorganisms offer a sustainable strategy for reducing N_2O emissions [1], since they mediate key processes such as nitrogen fixation, the conversion of atmospheric nitrogen into bioavailable ammonia, and nitrification, the oxidation of ammonia into nitrate, both of which influence nitrogen availability and greenhouse gas fluxes [3]. However, single strains are ineffective at industrial scale, requiring the design of microbial communities for function and safety [4]. Current approaches to designing microbial consortia largely rely on metagenomic data, limiting predictive power and ecological generalizability [5].

Prior work has developed ecological and computational frameworks for assembling high-performing consortia, often using genetic and evolutionary algorithms to navigate combinatorial complexity [6], and constraint-based modeling tools for metabolic and network inference [4]. Separately, machine learning (ML) and deep learning (DL) approaches have been applied to high-dimensional microbiome datasets [5]. Lastly, recent AI-assisted approaches to synthetic microbial community design demonstrate the ability to tailor communities to specific environmental contexts [7].

Despite these advances, there remains no framework that combines AI and mechanistic representations while simultaneously accounting for ecological dynamics, region-specific adaptation, and practical software accessibility. This thesis addresses that gap by developing a hybrid mechanistic–ML framework, in which mechanistic models are explicitly integrated into the learning process to guide predictions under sparse and heterogeneous real-world conditions. Combining this integration strategy with multi-modal microbiome data and open-access software, enables robust, interpretable, and regionally adaptive microbial community design while reducing dependence on nitrogen fertilizers. My thesis is guided by the following research questions:

1. How can mechanistic ecological simulations be leveraged to generate high-fidelity synthetic microbiome datasets, and what validation frameworks can ensure their statistical, structural, and functional alignment with real-world observations?

2. How can mechanistic models be embedded within machine learning frameworks to improve generalization and interpretability under sparse and heterogeneous data conditions?
3. How can hybrid predictive models be translated into region-aware decision support tools that incorporate environmental constraints, and are deployable as accessible, open-source software?

2 Research Plan and Timeline

The first aim of this research is to obtain the data needed. To achieve this, I will integrate microbial genomic, metabolic, physiological, and environmental data from public databases, ensuring respect for data ownership and sovereignty, with a focus on including diverse ecoregions, particularly underrepresented ecosystems in Africa and South America. Mechanistic models will be employed to generate synthetic data to supplement sparse or incomplete datasets, with synthetic labels explicitly indicated to avoid inflating performance estimates. Statistical tests and consistency checks will be used to ensure that the synthetic data generated obeys biological and physical constraints while preserving relationships between variables such as gene-gene interactions. Sim2Real validation techniques, including quantifying the gap between simulation and reality, will be used to ensure that the resulting framework learns true biological dynamics.

The second aim builds the predictive framework capable of identifying microbial communities that efficiently deliver nitrogen to plants while minimizing N₂O emissions. Advanced AI techniques, including tree ensembles for candidate taxa identification, representation learning and graph neural networks for modeling community interactions, and Bayesian methods for uncertainty quantification, will predict high-performing microbial communities. Multi-objective optimization will evaluate candidate communities based on biological fit, ecological stability, and efficiency, resulting in insights microbial consortia design and a corresponding publication detailing the framework.

Building on Aim 2, the third aim focuses on software development. Environmental recommendation will be added to the predictive framework, ensuring that the predicted microbial communities are tailored to specific ecoregions, then a modular, accessible software that generalizes the framework for *in silico* microbial community design will be created. ML models, including generalized additive models to capture non-linear relationships, artificial neural networks to predict habitat suitability, ensemble models to reduce uncertainty, and reinforcement learning for multi-objective optimization, will balance competing goals of stability, efficiency, and ecological appropriateness. The software platform will allow user-defined datasets, provide visualization modules for community interactions, metabolic fluxes, and ecological stability, and ensure interoperability with existing omics and soil data formats. Rigorous internal testing and external collaborations will validate reproducibility and robustness. This will result in an open-source research tool that provides region-specific microbial community recommendations, accompanied by a publication describing the design, functionality, and applications of the platform, thereby democratizing access to AI-driven microbial community design for researchers across diverse ecological and data-limited contexts.

Table 1. Expected timeline of proposed dissertation.

Period	Milestone
2026.9 – 2027.6	Data collection and synthetic data generation
2027.6 – 2028.1	Predictive framework development and validation
2028.1 – 2028.12	Software development and validation
2028.12 – 2029.6	Dissertation writing and defense preparation

3 Anticipated Contributions and Challenges

This research will pioneer an integrative framework that unites AI, systems biology, and ecological modeling to address the twin challenges of sustainable agriculture and climate change mitigation. By enabling rational, data-driven design of nitrogen-fixing and nitrifying microbial consortia, this research will reduce dependence on nitrogen fertilizers and mitigate agricultural greenhouse emissions, provide open and equitable access to advanced computational tools for global researchers, create a scalable model applicable to other biogeochemical and industrial systems (e.g., carbon sequestration, waste valorization), and introduce novel hybrid AI–mechanistic modeling strategies that explicitly handle uncertainty, sparsity, and ecological heterogeneity.

The primary challenges in this research include (i) severe data sparsity and imbalance across ecoregions, (ii) integrating mechanistically simulated data with real-world observations without inflating performance estimates, (iii) modeling complex, higher-order microbial interactions, and (iv) ensuring interpretability and uncertainty quantification in high-dimensional, multimodal models.

References

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